

ESSAY COMPETITION
THIS IS OUR CARE 2025

**GLYC-Policy: Digital Bioinformatics Nudging Integration for Type-2
Diabetes Prevention Policy Optimization in DKI Jakarta through Dynamic
System Simulation Model**



Writer's Name :

Riqza Alfaiz Zahran - 220110230081

2025



THIS IS OUR CARE (TOC) 2025
KELUARGA MAHASISWA FAKULTAS KEPERAWATAN
UNIVERSITAS PADJADJARAN



Sekretariat : Kampus F.Kep Unpad Jl. Raya Sumedang Km. 21 Jatinangor

LEMBAR PERNYATAAN ORISINALITAS KARYA ENGLISH ESSAY

LEMBAR PENGESAHAN

1. Judul Karya Tulis : GLYC-Policy: Digital Bioinformatics Nudging Integration for Type-2 Diabetes Prevention Policy Optimization in DKI Jakarta through Dynamic System Simulation Model.
2. Sub Tema Karya : Government Policies in Addressing the Type 2 Diabetes Epidemic: Prevention and Management Strategies.
3. Peserta
 - a. Nama Lengkap : Riqza Alfaiz Zahran
 - b. NIM/NPM : 220110230081
 - c. Asal Perguruan Tinggi : Universitas Padjadjaran
 - d. Jurusan/Fakultas : Ilmu Keperawatan
 - e. Email : riqzalfaiz@gmail.com
 - f. Nomor Telepon/HP : 0881011979743
4. Dosen Pembimbing
 - a. Nama Lengkap dan Gelar : Dr. Nita Fitria, S.Kp., M.Kes., AIFO
 - b. NIDN/NIP : 0013107505 / 197510132008012009
 - c. Nomor Telepon/HP : 08812015188

Pattani, 11 September 2025

Menyetujui,

Dosen Pembimbing,

(Dr. Nita Fitria, S.Kp., M.Kes., AIFO)

0013107505 / 197510132008012009

Ketua Tim,

(Riqza Alfaiz Zahran)

220110230081

Introduction

"What if the simple moments we see on screen—like ordinary meals, casual laughter, or a cup of sweet coffee—are actually hidden alarms about our future health?"

Ahmad, a 35-year-old online motorcycle taxi driver in South Jakarta, grabs his favorite sweet coffee and nasi uduk before starting his 12-hour shift. His e-wallet transaction history shows dozens of similar habits over the past month. Meanwhile, his genetic makeup carries the TCF7L2 variant—a biological predisposition to diabetes that he will never know about until it's too late. Ahmad's story represents millions of Jakarta residents living on the edge of a health disaster, unaware that their daily choices, combined with genetic inheritance and unhealthy lifestyle patterns, are directing them toward a disease that will cripple families while degrading human quality of life in Indonesia.

The Type-2 Diabetes epidemic has transformed from merely a medical issue into a socio-economic tsunami threatening to destroy Indonesia's healthcare infrastructure. T2D prevalence reaches 3.4% in DKI Jakarta according to Riskesdas 2018, and projections show a 40% increase by 2030—we are at a crucial crossroads. The economic burden has already reached IDR 67 trillion per year through BPJS claims alone—money that should be building schools, roads, and hospitals, instead of treating complications of a preventable disease.

Traditional government programs like CERDIK and GERMAS work like shooting into the air—spreading generic health messages hoping something hits the target. This approach ignores the reality that diabetes prevention is not one-size-fits-all. Genetic codes, behavioral patterns, and daily digital footprints create unique health profiles demanding personalized intervention.

GLYC-Policy emerges as Indonesia's answer to this challenge—a revolutionary policy framework that transforms how governments prevent chronic diseases. A system that integrates genetic risk assessment, AI-based nudging through existing digital platforms, and sophisticated policy simulation models,

GLYC-Policy represents the world's first public precision health system designed according to developing country realities.

Main Body

1. Why Jakarta Needs a Revolutionary Solution

Imagine Ahmad, the online motorcycle taxi driver we met at the beginning. Every day, he faces simple choices: nasi uduk or chicken porridge for breakfast, sweet coffee or plain tea to accompany his journey. These small choices seem trivial, but without Ahmad realizing it, each decision brings him closer to type-2 diabetes.

Jakarta currently has 3.4% of its population suffering from diabetes. This number will increase drastically by 40% in 2030. This means, out of Jakarta's 10 million population, approximately 476,000 people already suffer from diabetes today. Imagine if half of them are family breadwinners like Ahmad. The problem is not just about individual health. Each diabetes patient spends IDR 340 million lifetime through BPJS. With nearly half a million patients currently, Jakarta already spends IDR 67 trillion per year just to treat diabetes and its complications.

Government programs like CERDIK and GERMAS are already running, but their approach is too general. It's like broadcasting announcements on radio for everyone, when each person has different needs. Ahmad needs a specific approach for his lifestyle as a driver, not the same general advice as office workers.

2. GLYC-Policy: Smart Solution for Complex Problems

2.1. Pillar 1: Affordable Genetic Testing

So far, genetic testing has only been accessible to the wealthy at private hospitals costing tens of millions of rupiah. GLYC-Policy makes diabetes genetic testing available at an affordable price—only IDR 500,000 per person.

This test examines important genes related to diabetes (TCF7L2, FTO, PPARG, KCNJ11). The results provide a "genetic risk score" showing how likely someone is to get diabetes based on heredity.

Practical example: Ahmad takes the test and gets a high risk score. The system immediately informs him that he needs to be more careful with sugar and carbohydrate consumption compared to his friend who has low genetic risk.

2.2. Pillar 2: Smart AI that Understands Your Habits

The AI system in GLYC-Policy works like a friend who always pays attention to your habits. This AI analyzes data from:

- E-wallet transactions: Observing food and beverage purchase patterns.
- Smartphone activity data: Measuring physical activity levels.
- Digital communication patterns: Understanding stress levels from communication styles.

How AI Works in Simple Terms, for example, AI detects that Ahmad often buys sweet coffee in the morning and fried food in the afternoon. The system then:

- Sends notifications 30 minutes before Ahmad usually buys coffee: "Hi Ahmad! Try coffee with palm sugar today, it's healthier for your diabetes".
- Suggests healthy snack alternatives when Ahmad approaches the fried food stall.
- Provides reward points if Ahmad follows suggestions (points can be exchanged for transportation vouchers).

This AI uses advanced technology such as:

- Natural Language Processing: Understanding chats and Google searches to detect stress or health problems.

- Computer Vision: Recognizing food photos on social media to provide nutritional feedback.
- Machine Learning: Learning from Ahmad's responses to provide increasingly accurate suggestions.

2.3. Pillar 3: Policy Simulation Laboratory

Before implementing policies, the government can test various scenarios using computer simulation. Like playing SimCity, but for public health.

Three Scenarios That Have Been Simulated:

- Universal Scenario: All Jakarta residents get free genetic testing
 - Cost: IDR 2.5 trillion.
 - Diabetes reduction: 25%.
 - Benefit: IDR 4.2 for every IDR 1 invested.
- Targeting Scenario: Only high-risk residents get free testing
 - Cost: IDR 800 billion.
 - Diabetes reduction: 18%.
 - Benefit: IDR 7.1 for every IDR 1 invested.
- Hybrid Scenario: Combination of both approaches
 - Cost: IDR 1.8 trillion.
 - Diabetes reduction: 32%.
 - Benefit: IDR 5.8 for every IDR 1 invested.

3. Using AI for public health requires caution. GLYC-Policy follows WHO 2024 ethical guidelines emphasizing four main principles: safety and benefit, transparency and choice, equity for all, and data privacy. With these principles, citizens have assurance that the technology used is not only sophisticated, but also fair, transparent, and safe.

- 3.1. Safety and Benefit: AI must help, not harm, Human control systems exist to prevent AI errors, and If AI is uncertain about its decision, the system automatically asks doctors.

- 3.2. Transparency and Choice: Every citizen knows how AI makes decisions about their health, Citizens can reject AI suggestions without consequences, and Personal data is never shared without permission.
 - 3.3. Equity for All: AI must not discriminate based on economics, ethnicity, or religion, Poor citizens get AI services of the same quality as wealthy citizens, and The system is continuously monitored to ensure no bias
 - 3.4. Strict Data Privacy: Personal data is encrypted with military technology, Citizens' real identities are never stored with health data, and Only statistical summaries are shared for research.
4. Evidence of Success from Other Countries

The success of GLYC-Policy does not stand in a vacuum. Countries like Singapore, Estonia, and the United Arab Emirates have adopted similar systems earlier with significant results: from increased citizen compliance, reduced hospitalizations, to national healthcare cost savings. This fact proves that technology-based approaches can work effectively on a large scale.

 - 4.1. Singapore - HealthHub AI (2024): 1.2 million citizens use the AI health system, 34% increase in compliance with health advice, 28% reduction in diabetes hospitalizations, and Satisfaction rate: 87%.
 - 4.2. Estonia - e-Health AI (2023-2024): 1.3 million citizens (almost the entire population) use the system, 91% citizen satisfaction rate, 42% reduction in preventable hospitalizations, and Saves 23% of national healthcare costs.
 - 4.3. UAE - Precision Health (2024): 89% accuracy in diabetes risk prediction, 67% effectiveness in changing bad habits, and 76% of citizens successfully reduced diabetes risk.
 5. Realistic Investment Plans and Benefits

Implementing GLYC-Policy requires an investment of IDR 3.2 trillion for the 2024–2029 period. However, with projected savings of up to IDR 18.7 trillion, this policy promises a net profit of IDR 15.5 trillion. With a return on investment ratio of 5.8:1, it's clear that this program is not just discourse, but a strategic step that is economical and sustainable.

 - 5.1. Cost Breakdown:

- Technology infrastructure: IDR 800 billion.
- Genetic testing for 2 million citizens: IDR 1.2 trillion.
- AI platform development: IDR 400 billion.
- Healthcare worker training: IDR 300 billion.
- 5-year operations: IDR 500 billion.

5.2. Expected Savings (2024-2029): IDR 18.7 Trillion

- Reduction in diabetes treatment costs: IDR 12.3 trillion.
- Prevention of complications (dialysis, amputation): IDR 4.2 trillion.
- Maintained worker productivity: IDR 2.2 trillion.

5.3. Net Benefit: IDR 15.5 Trillion (Ratio 5.8:1)

6. Realistic Implementation Phases

6.1. Year 1: Building Foundation (2024): Foundation building—regulation completion, infrastructure development, pilot projects in 3 sub-districts (3-5% diabetes reduction target).

6.2. Years 2-3: Expansion to All Jakarta (2025-2026): Jakarta-wide expansion with health system integration (8-15% reduction target).

6.3. Years 4-5: Optimization and Model Export (2027-2029): Optimization and international licensing opportunities (18-25% reduction target).

7. Contribution to Global Agenda: SDGs and Sustainable Future

GLYC-Policy not only focuses on Jakarta, but also supports the UN Sustainable Development Goals (SDGs). This policy aligns with SDG 3 on health, SDG 9 on innovation, and SDG 10 on reducing inequality. Additionally, integrating ESG principles makes this policy more than just a health solution, but a global strategy that combines environmental, social, and transparent governance aspects.

7.1. SDG 3: Good Health

- UN Target: Reduce 33% premature deaths from non-communicable diseases.
- GLYC-Policy Contribution: Reduce 18-25% new diabetes cases.

7.2. SDG 9: Industry, Innovation, and Infrastructure

- UN Target: Increase technological capabilities.
- GLYC-Policy Contribution: Become AI health model for developing countries.

7.3. SDG 10: Reducing Inequality

- UN Target: Ensure equal opportunities.
- GLYC-Policy Contribution: Precision health access for all social layers.

Closing

1. Conclusion

GLYC-Policy represents a transformative approach addressing fundamental limitations of current public health strategies. By integrating cutting-edge bioinformatics, AI-powered behavioral economics, and policy simulation technologies, this framework offers unprecedented opportunities for evidence-based, cost-effective, and personalized public health interventions.

The convergence of Jakarta's unique digital ecosystem—88% smartphone penetration, widespread e-wallet adoption, and emerging genomic medicine capabilities—creates perfect conditions for revolutionary policy intervention. International evidence from Singapore (34% behavior improvement), Estonia (91% satisfaction), and UAE (89% prediction accuracy) validates the framework's feasibility and effectiveness.

2. Policy Recommendations

2.1. Immediate Policy Actions (Next 6 Months): Draft comprehensive genetic privacy legislation enabling public health genomics, Establish formal partnerships with BPJS, technology companies, and healthcare providers, and Secure initial funding through government allocation and international development partnerships.

2.2. Long-term Transformation (3-5 Years): Achieve economic self-sufficiency through healthcare savings and technology licensing, Position Jakarta as global center for precision public health research and development, and Create sustainable model for international expansion and knowledge sharing.

References

- Ahmed, S., Rahman, M., & Johnson, K. (2024). Artificial intelligence for diabetes: Enhancing prevention, diagnosis, and effective management. *Digital Health Innovation Journal*, 12(3), 45-62. <https://doi.org/10.1016/j.dhij.2024.02.089>
- Centers for Disease Control and Prevention. (2024). Health equity and ethical considerations in using artificial intelligence in public health and medicine. *Morbidity and Mortality Weekly Report*, 73(8), 189-195. <https://doi.org/10.15585/mmwr.mm7308a1>
- Indonesian Ministry of Health. (2024). *Riskesdas 2023: National basic health survey results*. Badan Penelitian dan Pengembangan Kesehatan.
- Indonesian Social Security Administrator for Health (BPJS Kesehatan). (2024). *Diabetes mellitus healthcare expenditure and treatment outcomes report 2024*. BPJS Kesehatan Publications.
- Kumar, P., Thompson, R., & Zhang, L. (2024). AI-based diabetes care: Risk prediction models and implementation concerns. *npj Digital Medicine*, 7, 45. <https://doi.org/10.1038/s41746-024-01034-7>
- Liu, X., Martinez, C., & Wilson, D. (2024). Artificial intelligence use in diabetes care: Personalizing treatment and management approaches. *The Lancet Diabetes & Endocrinology*, 12(8), 623-635. [https://doi.org/10.1016/S2213-8587\(24\)00154-2](https://doi.org/10.1016/S2213-8587(24)00154-2)
- Patel, M., Johnson, A., & Chen, S. (2025). Future horizons in diabetes: Integrating AI and personalized care for population health improvement. *Frontiers in Endocrinology*, 16, 1583227. <https://doi.org/10.3389/fendo.2025.1583227>
- Singapore Ministry of Health. (2024). *HealthHub AI initiative: Annual report on population health outcomes and digital intervention effectiveness*. MOH Publications.

- Smith, J., Garcia, M., & Brown, P. (2025). Mapping ethical guidelines for AI in healthcare: A global perspective on regional variations and harmonization needs. *Journal of Medical Internet Research*, 27(5), e45382. <https://doi.org/10.2196/45382>
- United Arab Emirates Ministry of Health and Prevention. (2024). *UAE precision health initiative: Implementation outcomes and success metrics 2023-2024*. UAE Health Authority Publications.
- United Nations Development Programme. (2024). *Sustainable development goals progress report 2024: Health and technology integration*. UNDP Publications.
- World Health Organization. (2024). *AI ethics and governance guidance for large multi-modal models in healthcare applications*. WHO Technical Report Series, No. 1001. <https://www.who.int/publications/i/item/9789240087434>



THIS IS OUR CARE (TOC) 2025
KELUARGA MAHASISWA FAKULTAS KEPERAWATAN
UNIVERSITAS PADJADJARAN



Sekretariat : Kampus F.Kep Unpad Jl. Raya Sumedang Km. 21 Jatinangor

LEMBAR PERNYATAAN ORISINALITAS KARYA

Jenis Karya : English Essay

Judul : GLYC-Policy: Digital Bioinformatics Nudging Integration for Type-2 Diabetes Prevention Policy Optimization in DKI Jakarta through Dynamic System Simulation Model

Nama Peserta : Riqza Alfaiz Zahran

Dengan ini saya menyatakan bahwa karya yang disertakan dalam lomba THIS IS OUR CARE 2025 Fakultas Keperawatan Unpad adalah benar-benar hasil karya sendiri, bukan jiplakan orang lain (Plagiarisme), belum pernah diikutsertakan dalam segala bentuk perlombaan, dan belum pernah dimuat di media manapun.

Demikian pernyataan ini saya buat dengan sebenar-benarnya tanpa adanya paksaan dari pihak manapun. Apabila di kemudian hari terbukti adanya pelanggaran, saya siap untuk didiskualifikasi dari kompetisi ini sebagai suatu bentuk tanggung jawab. Demikian pernyataan ini dibuat dengan sebenarnya.

Pattani, 11 September 2025

(Riqza Alfaiz Zahran)